

Product Requirements Document (PRD)

Full Inventory Management System

Version: 1.0 **Prepared for:** Inventory Management SaaS/Web App **Tech Stack:** MERN (MongoDB, Express.js, React.js, Node.js) + Tailwind CSS + GSAP + Lucide Icons

1. Overview / Introduction

Yeh document ek **Full Inventory Management System** ke liye Product Requirements define karta hai. System ka maqsad hai kisi bhi business (retail, wholesale, warehouse, manufacturing) ke liye stock, purchases, sales, suppliers aur reporting ko ek modern, animated, aur professional-look wale dashboard se manage karna.

App ka focus:

Logical & scalable architecture (MERN stack)

Smooth animations (GSAP) taake UX premium feel de

Clean, modern UI (Tailwind CSS + Lucide Icons)

Real-world business logic (multi-warehouse, low-stock alerts, purchase/sale orders, reports)

2. Goals & Objectives

Goal	Description
Centralized Stock Control	Ek hi jagah se saare products, categories aur stock levels track hon
Real-time Alerts	Low stock, expiry, aur reorder alerts automatically milen
Multi-Role Access	Admin, Manager, Staff ke alag permissions
Sales & Purchase Tracking	Purchase orders aur sales orders ka complete lifecycle
Reporting & Analytics	Visual charts se business insights milein
Modern UX	Animated transitions, micro-interactions, responsive design

3. Target Users

Admin – Full control (users, settings, reports, all modules)
Manager – Inventory, purchase, sales management
Staff/Cashier – Sales entry, stock check
Supplier (optional portal) – Purchase order status dekh sake

4. Core Features (Modules)

4.1 Authentication & User Management

JWT-based login/signup
Role-based access control (RBAC): Admin / Manager / Staff
Password reset via email (Nodemailer)
Profile management with avatar upload
Activity logs (kis user ne kya action kiya)

4.2 Dashboard

Animated stat cards (Total Products, Total Stock Value, Low Stock Items, Today's Sales)
Sales vs Purchase trend chart (line/bar chart with animation)
Recent activity feed
Quick action buttons (Add Product, New Sale, New Purchase)

4.3 Product Management

Add/Edit/Delete products
Product fields: Name, SKU, Barcode, Category, Brand, Unit, Cost Price, Selling Price, Tax, Stock Qty, Reorder Level, Images
Bulk import/export (CSV/Excel)
Barcode generation & scanning support
Product variants (size, color, etc.)

4.4 Category & Brand Management

CRUD for categories & sub-categories
Brand management
Drag-and-drop category reordering (animated)

4.5 Stock / Warehouse Management

Multi-warehouse / multi-location support
Stock transfer between warehouses
Stock adjustment (damage, loss, correction) with reason logging
Real-time stock level indicators (color-coded: green/yellow/red)

4.6 Purchase Management

Create Purchase Orders (PO) to suppliers
Receive stock against PO (partial/full receiving)
Purchase return handling
Supplier ledger/history

4.7 Sales Management

POS-style sales screen (fast add-to-cart UI)
Invoice generation (PDF export)
Sales return / refund handling
Customer management (basic CRM: name, contact, purchase history)
Discounts, taxes, multiple payment methods

4.8 Supplier Management

Supplier CRUD
Supplier performance tracking (on-time delivery, pricing history)

4.9 Reports & Analytics

Stock valuation report
Sales report (daily/weekly/monthly)
Profit & Loss summary
Low stock / out-of-stock report
Best-selling products report
Export to PDF/Excel
Animated charts (Recharts/Chart.js with GSAP transitions)

4.10 Notifications & Alerts

Low stock alerts (in-app + email)
Expiry date alerts (for perishable inventory)
Order status notifications
Toast notifications with smooth animations

4.11 Settings

Company profile & branding
Tax settings
Currency & unit settings
Theme toggle (Light/Dark) with animated transition
Backup/export data

5. Non-Functional Requirements

Requirement	Detail
Performance	Page load < 2s, API response < 300ms
Scalability	Support 100k+ products, multi-tenant ready
Security	JWT, bcrypt password hashing, input validation, rate limiting
Responsiveness	Fully mobile/tablet/desktop responsive
Accessibility	Keyboard navigation, ARIA labels
Availability	99.9% uptime target

6. UI/UX & Animation Requirements

Since **well-designed animated theme** requested hai, yahan specific guidance hai:

Design System

Tailwind CSS for utility-first styling, custom design tokens (colors, spacing, radius)

Lucide Icons for all icons (consistent, lightweight, tree-shakable)

Dark/Light mode with animated toggle (theme colors transition smoothly)

Card-based layout with soft shadows & rounded corners

Consistent color palette: Primary (brand color), Success (green), Warning (yellow), Danger (red)

GSAP Animation Use-Cases

Element	Animation
Page transitions	Fade + slide on route change
Dashboard cards	Stagger fade-in on load
Sidebar menu	Smooth expand/collapse
Modals (Add/Edit forms)	Scale + fade entrance
Toast notifications	Slide-in from top-right

Element	Animation
Charts	Animated bar/line growth on render
Stock level bars	Animated progress bar fill
Buttons	Hover micro-interactions (scale, ripple)
Loading states	Skeleton loaders with shimmer effect
Drag & drop (kanban-style order status) Smooth reordering animation	

7. Technical Architecture

Tech Stack

Frontend: React.js (Vite), Tailwind CSS, GSAP, Lucide React, Recharts/Chart.js, React Router, Redux Toolkit / Zustand (state management), Axios

Backend: Node.js, Express.js, JWT Auth, Multer (file uploads), Nodemailer

Database: MongoDB (Mongoose ODM)

Other: Socket.io (real-time stock updates/notifications), PDFKit/jsPDF (invoice/report generation)

High-Level Folder Structure

inventory-system/ client/ (React + Tailwind + GSAP) src/ components/ pages/ hooks/ store/ animations/ (GSAP timelines) utils/ server/ (Node + Express) models/ routes/ controllers/ middleware/ utils/ shared/ (types/constants)

Core Database Models (MongoDB Collections)

User	(name, email, password, role, avatar)
Product	(name, sku, barcode, category, brand, price, cost, stockQty, reorderLevel, warehouse)
Category	
Brand	
Warehouse	
Supplier	
PurchaseOrder	(supplier, items[], status, totalAmount)
SalesOrder	(customer, items[], status, totalAmount, payment)
Customer	
StockMovement	(type: in/out/transfer/adjustment, product, qty, reason)
ActivityLog	

Sample REST API Endpoints



```
POST /api/auth/login
POST /api/auth/register
GET /api/products
POST /api/products
PUT /api/products/:id
DELETE /api/products/:id
GET /api/stock/movements
POST /api/purchase-orders
POST /api/sales-orders
GET /api/reports/sales
GET /api/reports/stock-valuation
GET /api/dashboard/stats
```

8. Recommended Hosting

Aapke MERN + real-time (Socket.io) requirements ko dekhte hue, yeh recommendation hai:

Component	Recommended Service	Why
Frontend (React build)	Vercel or Netlify	Free tier, fast CDN, auto CI/CD from GitHub, perfect for Vite/React apps
Backend (Node/Express)	Render or Railway	Easy Node.js deployment, free/cheap tier, auto-deploy from Git, supports WebSockets (for Socket.io)
Database (MongoDB)	MongoDB Atlas	Managed MongoDB, free M0 tier for small apps, easy scaling, built-in backups
File/Image Storage	Cloudinary	Free tier, optimized image delivery, easy Multer integration
Email Service	Resend or SendGrid	For alerts/notifications (low stock, password reset)
Domain + SSL	Namecheap/GoDaddy + Cloudflare	Cheap domain + free SSL & CDN via Cloudflare

Recommended Setup for Production-Scale (if traffic grows)

Backend: AWS EC2 / DigitalOcean Droplet + PM2 + Nginx (reverse proxy)

Database: MongoDB Atlas (Dedicated cluster)

Load Balancing: AWS ALB or Nginx

File Storage: AWS S3 + CloudFront

Monitoring: UptimeRobot / Sentry (error tracking)

Simple recommendation for MVP/Startup budget:

Frontend → **Vercel** | Backend → **Render** | Database → **MongoDB Atlas (Free M0)** | Images → **Cloudinary**

Yeh combination free/low-cost hai, easily scalable hai, aur MERN + Socket.io ke sath best compatibility rakhta hai.

9. Development Phases (Suggested Roadmap)

Phase	Scope	Est. Duration
Phase 1	Auth, Product/Category CRUD, Basic Dashboard	2 weeks
Phase 2	Stock management, Warehouse, Purchase Orders	2 weeks
Phase 3	Sales/POS module, Invoicing, Customers	2-3 weeks

Phase	Scope	Est. Duration
Phase 4	Reports, Analytics, Notifications	1-2 weeks
Phase 5	GSAP Animations, UI Polish, Testing, Deployment	1-2 weeks

10. Success Metrics

90%+ inventory accuracy vs physical stock
< 2 sec dashboard load time
95%+ user satisfaction on UI/UX (animation & ease of use)
Zero critical bugs post-launch (first 30 days)

Is PRD ko aap apni team/developer ke sath share kar sakte hain scope, design aur development plan finalize karne ke liye.